

```

    return 0;
}
manchodu@Srinu:~/OSLAB$ nano 52.c
manchodu@Srinu:~/OSLAB$ gcc m52.c -o m52
cc1: fatal error: m52.c: No such file or directory
compilation terminated.
manchodu@Srinu:~/OSLAB$ cat m52.c
cat: m52.c: No such file or directory
manchodu@Srinu:~/OSLAB$ gcc 52.c -o 52
52.c:1:3: error: invalid preprocessing directive #Makefile
1 | # Makefile
  | ^~~~~~
52.c:3:3: error: invalid preprocessing directive #Build
3 | # Build all shell project modules
  | ^~~~~~
52.c:5:3: error: invalid preprocessing directive #Commands
5 | # Commands:
  | ^~~~~~
52.c:6:3: error: invalid preprocessing directive #make
6 | # make
  | ^~~~~~
52.c:7:3: error: invalid preprocessing directive #make
7 | # make clean
  | ^~~~~~
52.c:8:3: error: invalid preprocessing directive #make
8 | # make test
  | ^~~~~~
52.c:9:3: error: invalid preprocessing directive #make
9 | # make valgrind
  | ^~~~~~
52.c:10:3: error: invalid preprocessing directive #make
10 | # make run-pipeline
   | ^~~~~~
52.c:11:3: error: invalid preprocessing directive #make
11 | # make run-signals
   | ^~~~~~
52.c:13:1: warning: data definition has no type or storage class
13 | CC = gcc
   | ^~~
52.c:13:1: warning: type defaults to 'int' in declaration of 'CC' [-Wimplicit-int]
52.c:13:6: error: 'gcc' undeclared here (not in a function)
13 | CC = gcc
   | ^~~~
52.c:15:1: error: expected ',' or ';' before 'CFLAGS'
15 | CFLAGS = -Wall -Wextra -g
   | ^~~~~~
52.c:36:32: error: stray '@' in program
36 | $(CC) $(CFLAGS) $< -o $@
   |                                ^
52.c:39:32: error: stray '@' in program
39 | $(CC) $(CFLAGS) $< -o $@
   |                                ^
52.c:42:32: error: stray '@' in program
42 | $(CC) $(CFLAGS) $< -o $@
   |                                ^
52.c:45:32: error: stray '@' in program
45 | $(CC) $(CFLAGS) $< -o $@
   |                                ^
52.c:48:32: error: stray '@' in program
48 | $(CC) $(CFLAGS) $< -o $@
   |                                ^
52.c:51:32: error: stray '@' in program
51 | $(CC) $(CFLAGS) $< -o $@
   |                                ^
52.c:54:32: error: stray '@' in program
54 | $(CC) $(CFLAGS) $< -o $@
   |                                ^
52.c:57:32: error: stray '@' in program
57 | $(CC) $(CFLAGS) $< -o $@
   |                                ^
52.c:60:32: error: stray '@' in program
60 | $(CC) $(CFLAGS) $< -o $@
   |                                ^

```

```

anchodu@Srin:~/OSLAB$ cat 52.c
Makefile

Build all shell project modules

Commands:
make
make clean
make test
make valgrind
make run-pipeline
make run-signals

C = gcc

FLAGS = -Wall -Wextra -g

BIN = bin

PROGRAMS = \
    $(BIN)/integrated_shell_modules \
    $(BIN)/signal_handler_management \
    $(BIN)/foreground_interrupt_handling \
    $(BIN)/suspend_resume_jobs \
    $(BIN)/process_group_terminal_control \
    $(BIN)/pipeline_stability \
    $(BIN)/pipeline_monitor \
    $(BIN)/memory_cleanup_test \
    $(BIN)/error_scenario_testing

all: $(BIN) $(PROGRAMS)

$(BIN):
    mkdir -p $(BIN)

$(BIN)/integrated_shell_modules: src/integrated_shell_modules.c
    $(CC) $(CFLAGS) $< -o $@

$(BIN)/signal_handler_management: src/signal_handler_management.c
    $(CC) $(CFLAGS) $< -o $@

$(BIN)/foreground_interrupt_handling: src/foreground_interrupt_handling.c
    $(CC) $(CFLAGS) $< -o $@

$(BIN)/suspend_resume_jobs: src/suspend_resume_jobs.c
    $(CC) $(CFLAGS) $< -o $@

$(BIN)/process_group_terminal_control: src/process_group_terminal_control.c
    $(CC) $(CFLAGS) $< -o $@

$(BIN)/pipeline_stability: src/pipeline_stability.c
    $(CC) $(CFLAGS) $< -o $@

$(BIN)/pipeline_monitor: src/pipeline_monitor.c
    $(CC) $(CFLAGS) $< -o $@

$(BIN)/memory_cleanup_test: src/memory_cleanup_test.c
    $(CC) $(CFLAGS) $< -o $@

$(BIN)/error_scenario_testing: src/error_scenario_testing.c
    $(CC) $(CFLAGS) $< -o $@

test: all
    chmod +x tests/test_shell.sh
    chmod +x tests/run_all_tests.sh
    ./tests/test_shell.sh
    ./tests/run_all_tests.sh

valgrind: $(BIN)/memory_cleanup_test
    valgrind --leak-check=full --show-leak-kinds=all ./$$(BIN)/memory_cleanup_test

run-pipeline: $(BIN)/pipeline_stability
    ./$$(BIN)/pipeline_stability

run-monitor: $(BIN)/pipeline_monitor
    ./$$(BIN)/pipeline_monitor

run-signals: $(BIN)/signal_handler_management
    ./$$(BIN)/signal_handler_management

run-jobs: $(BIN)/suspend_resume_jobs
    ./$$(BIN)/suspend_resume_jobs

run-shell: $(BIN)/integrated_shell_modules
    ./$$(BIN)/integrated_shell_modules

clean:

```

```
Run Jobs Run Shell
hanchodu@Srini:~/OSLAB$ # docs/syscalls.txt
```

SYSTEM CALLS USED

```
Fork()
Creates a new process.

execvp()
Replaces the current process with another program.

waitpid()
Waits for child process state changes.

pipe()
Creates an inter-process communication channel.

dup2()
Duplicates file descriptors for redirection.

open()
Opens files.

read()
Reads data from a file descriptor.

write()
Writes data to a file descriptor.

close()
Closes file descriptors.

kill()
Sends signals to processes or process groups.

sigaction()
Registers signal handlers.

setpgid()
Creates or assigns a process group.

getpgrp()
Returns the process group ID.

tcsetpgrp()
Transfers terminal control to a process group.

tcgetpgrp()
Sets the current terminal process group.

malloc()
Allocates dynamic memory.

free()
Releases dynamic memory.

getrusage()
Retrieves resource usage information.

clock_gettime()
Measures execution time.

chdir()
Changes the current working directory.

getcwd()
Gets the current working directory.
```